

Forward Orthogonal Chains and Hamel Dimension

FA/Banach automatic math run: `fa_banach_001`

2026-07-18

Status

Partial result likely valid. The source paper asks which normed-space properties can be decided from Birkhoff–James orthogonality alone, and then shows, among other things, that its projective di-orthograph detects finite dimension, smoothness, and strict convexity in the stated settings [1]. The result below strengthens the dimension-recognition part for smooth spaces: the graph determines not only whether the dimension is finite, but the exact Hamel dimension cardinal.

Source Question

The extracted question is Question 2.1 in arXiv:2109.09414, page 4:

2. MAIN RESULTS – PROPERTY RECOGNITION

The fundamental question which we address here is the following:

Question 2.1. Let $\mathcal{P}$ be a given property on a normed space. Can we decide, using BJ orthogonality alone, if a space has property $\mathcal{P}$ or not?

Of course, this is too general for any hope of a definite answer and we will restrict ourselves to three very specific properties: the dimension, the smoothness, and the rotundness.

Let us first show that by means of $\hat{\Gamma}(\mathcal{X})$ one can calculate the dimension of a normed space $\mathcal{X}$.

Proposition 2.2. *A normed space $\mathcal{X}$ over $\mathbb{F}$ is infinite-dimensional if and only if $\hat{\Gamma}(\mathcal{X})$ contains an infinite clique.*

Proof. See our recent paper [5, Corollary 3.1 and Theorem 3.5]. $\square$

Lemma 2.3. *Let $\mathcal{X}$ be a normed space over $\mathbb{F}$ and $n \geq 2$. Then the following statements are equivalent.*

- (i) *For any $k \in \{1, \dots, n-1\}$ and any k -tuple of lines $(x_1, \dots, x_k)$ satisfying $x_i \perp x_j$, $1 \leq i < j \leq k$, one can always find $x_{k+1} \in \hat{\Gamma}(\mathcal{X})$ with $x_i \perp x_{k+1}$, $1 \leq i \leq k$.*
- (ii) *For any $(n-1)$ -tuple of lines $(x_1, \dots, x_{n-1})$, one can always find $x_n \in \hat{\Gamma}(\mathcal{X})$ with $x_i \perp x_n$, $1 \leq i \leq n-1$.*
- (iii) $\dim \mathcal{X} \geq n$.

Proof. We will once again rely on the fact that $x \perp y$ if and only if $y \in \text{Ker } f_x$, where f_x is a supporting functional at x .

(iii) $\implies$ (ii). Choose an arbitrary $(n-1)$ -tuple of lines $(x_1, \dots, x_{n-1})$. Let f_i be any supporting functional at x_i , $i = 1, \dots, n-1$. Denote $\mathcal{X}_{n-1} = \bigcap_{i=1}^{n-1} \text{Ker } f_i$. Then $\text{codim } \mathcal{X}_{n-1} \leq n-1$ together with $\dim \mathcal{X} \geq n$ imply that the subspace $\mathcal{X}_{n-1} \subseteq \mathcal{X}$ contains at least one line, x_n . Clearly, $x_i \perp x_n$ for $i = 1, \dots, n-1$.

The source immediately notes that the question is too broad for a definite global answer and restricts to the dimension, smoothness, and rotundness. The present packet stays in that dimension-recognition lane.

Definitions

Let X be a normed space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. Recall that x is Birkhoff–James orthogonal to y , written $x \perp y$, if

$$\|x + \lambda y\| \geq \|x\| \quad (\lambda \in \mathbb{F}).$$

The projective di-orthograph $\hat{\Gamma}(X)$ has as vertices the one dimensional subspaces $[x] = \mathbb{F}x$, $x \neq 0$, and a directed edge $[x] \rightarrow [y]$ exactly when $x \perp y$. In a smooth normed space every nonzero x has a unique supporting functional f_x normalized by $\|f_x\| = 1$ and $f_x(x) = \|x\|$. The James criterion recalled in the source gives

$$x \perp y \iff f_x(y) = 0$$

in the smooth case.

Definition 1. A forward orthogonal chain in $\widehat{\Gamma}(X)$ is a well-ordered family $(v_\alpha)_{\alpha < \kappa}$ of distinct vertices such that

$$v_\alpha \rightarrow v_\beta \quad (\alpha < \beta < \kappa).$$

Let

$$\text{fo}(X) = \sup\{\kappa : \widehat{\Gamma}(X) \text{ contains a forward orthogonal chain of cardinal } \kappa\}.$$

Proof Intuition

In a smooth normed space, each vertex $v = [x]$ carries a single hyperplane $\ker f_x$ of outgoing neighbors. A forward chain therefore behaves like a triangular biorthogonal system: earlier supporting functionals vanish on later vectors. This triangularity forces every finite subset of the chain to be linearly independent, giving the upper bound by the Hamel dimension.

For the reverse inequality, run the same idea by transfinite recursion. Before stage $\alpha < \dim X$, fewer than $\dim X$ supporting hyperplanes have been chosen, so their intersection is still nonzero. Choosing the next vector in that intersection makes all earlier vertices point to the new one.

Theorem 1. Let X be a smooth normed space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. Then

$$\text{fo}(X) = \dim_{\mathbb{F}} X.$$

Consequently, among smooth normed spaces, the projective Birkhoff–James di-orthograph determines the full Hamel dimension cardinal.

Proof. The zero-dimensional case is empty, so assume $X \neq \{0\}$. Put $\kappa = \dim_{\mathbb{F}} X$.

First let $(v_\alpha)_{\alpha < \lambda}$ be a forward orthogonal chain. Choose nonzero representatives $x_\alpha \in v_\alpha$, and let f_α be the unique supporting functional at x_α . If $\alpha < \beta$, then $v_\alpha \rightarrow v_\beta$, hence $x_\alpha \perp x_\beta$, and smoothness gives

$$f_\alpha(x_\beta) = 0.$$

We claim that the family $(x_\alpha)_{\alpha < \lambda}$ is linearly independent. Indeed, take a finite relation

$$a_1 x_{\alpha_1} + \cdots + a_n x_{\alpha_n} = 0, \quad \alpha_1 < \cdots < \alpha_n.$$

Applying f_{α_1} gives

$$a_1 f_{\alpha_1}(x_{\alpha_1}) = 0,$$

because $f_{\alpha_1}(x_{\alpha_j}) = 0$ for $j > 1$. Since $f_{\alpha_1}(x_{\alpha_1}) = \|x_{\alpha_1}\| \neq 0$, we get $a_1 = 0$. Repeating the argument with $f_{\alpha_2}, \dots, f_{\alpha_n}$ yields all $a_j = 0$. Thus every forward orthogonal chain has cardinal at most $\dim X$, so $\text{fo}(X) \leq \kappa$.

For the reverse inequality, well-order a Hamel basis cardinal of X as κ . We construct by transfinite recursion vectors $(x_\alpha)_{\alpha < \kappa}$ and their supporting functionals $(f_\alpha)_{\alpha < \kappa}$ so that

$$f_\beta(x_\alpha) = 0 \quad (\beta < \alpha).$$

Suppose the construction has been completed for all $\beta < \alpha$. Since $|\alpha| < \kappa$, the intersection

$$M_\alpha = \bigcap_{\beta < \alpha} \ker f_\beta$$

has codimension at most $|\alpha| < \kappa$. Therefore $M_\alpha \neq \{0\}$. Choose $0 \neq x_\alpha \in M_\alpha$, and let f_α be the unique supporting functional at x_α . Then for every $\beta < \alpha$, $f_\beta(x_\alpha) = 0$, so $x_\beta \perp x_\alpha$. Thus the vertices $([x_\alpha])_{\alpha < \kappa}$ form a forward orthogonal chain of cardinal κ . Hence $\text{fo}(X) \geq \kappa$, completing the proof. $\square$

Limitations and Novelty Check

The theorem is not a full answer to Question 2.1 of the source paper; that question is intentionally broad. It also does not remove smoothness. Without smoothness, an edge $x \rightarrow y$ may be witnessed by one supporting functional at x and another edge $x \rightarrow z$ by a different supporting functional, so the triangular linear-independence argument above can fail.

The local run indexes were searched for arXiv:2109.09414, Birkhoff–James orthogonality, Hamel dimension, forward chains, transitive cliques, and dimension-cardinal language. No exact duplicate was found. A web search on 2026-07-18 found the later open-access follow-up *Birkhoff–James classification of norm’s properties* [2]. That paper advances finite-dimensional property recognition, including smooth points, maximal faces, supremum norms, and Radon-plane examples, but I did not find the forward-chain/Hamel-cardinal statement above.

Human Review Recommendation

Review as a compact partial theorem. The key point is the use of smoothness in both directions: it converts graph edges into equations against unique support functionals. A possible follow-up would be to determine whether a modified graph invariant can recover Hamel cardinality without smoothness.

References

- [1] Lj. Arambasic, A. Guterman, B. Kuzma, R. Rajic, and S. Zhilina. *What does Birkhoff–James orthogonality know about the norm?* arXiv:2109.09414, 2021.
- [2] A. Guterman, B. Kuzma, S. Singla, and S. Zhilina. *Birkhoff–James classification of norm’s properties*. Advances in Operator Theory 9, Article 43, 2024. <https://doi.org/10.1007/s43036-024-00321-0>.